Directed numbers: multiplying and dividing integers

A Cambridge insert: two signs make a third, and one rule covers both operations.

LESSON 5–7 | 3 × 40 MIN | MATEMATIKA 6, QOʻSHIMCHA MAVZU | S7 1.2

LESSON CLOCK

20' | 25' | 30' | 30' | 15'

Why a minus times a minus is a plus	20 min
The rule of signs	25 min
Dividing	30 min
Mixed expressions	30 min
Squares and powers	15 min

LEARNING OBJECTIVES

- Multiply and divide integers using the sign rule.
- Explain the rule with repeated addition and with a pattern.
- Evaluate expressions mixing all four operations on integers.
- Square a negative number correctly.

TEXTBOOK

National	pp. 15–20
Cambridge	Stage 7, pp. 14–18
Workbook	Exercise 1.2

01 Explanation

Why a minus times a minus is a plus

Start from repeated addition, then continue the pattern downwards.

PRODUCT	VALUE
$3 \cdot (-4)$	-12
$2 \cdot (-4)$	-8
$1 \cdot (-4)$	-4
$0 \cdot (-4)$	0
$(-1) \cdot (-4)$	4
$(-2) \cdot (-4)$	8

THE PATTERN DECIDES IT

Each row adds 4 to the one above. Continuing past zero forces $(-1) \cdot (-4) = 4$ — the rule is not an arbitrary convention but the only way to keep the arithmetic consistent.

The rule of signs

> $+ \cdot + = + \quad + \cdot - = - \quad - \cdot + = - \quad - \cdot - = +$

PRODUCT	SIGNS	ANSWER
$6 \cdot 7$	same	42
$6 \cdot (-7)$	different	-42
$(-6) \cdot 7$	different	-42
$(-6) \cdot (-7)$	same	42

COUNT THE MINUS SIGNS

An even number of negative factors gives a positive product; an odd number gives a negative one. $(-1)(-2)(-3)$ has three, so the answer is -6.

Dividing

Division follows the same rule, because it is multiplication by the reciprocal.

QUOTIENT	ANSWER	CHECK
$-12 \div 3$	-4	$3 \cdot (-4) = -12$
$12 \div (-3)$	-4	$(-3)(-4) = 12$
$-12 \div (-3)$	4	$(-3) \cdot 4 = -12$
$0 \div (-3)$	0	$(-3) \cdot 0 = 0$

DIVIDING BY ZERO IS STILL IMPOSSIBLE

$0 \div (-3)$ is 0, but $-3 \div 0$ has no answer at all: no number multiplied by 0 gives -3. Negative numbers change nothing about that.

Mixed expressions

EXPRESSION	WORKING	ANSWER
$-3 \cdot 4 + 5$	$-12 + 5$	-7
$-3 \cdot (4 + 5)$	$-3 \cdot 9$	-27
$20 \div (-4) - 3$	$-5 - 3$	-8
$(-2)(-3)(-4)$	three minus signs	-24

THE ORDER OF OPERATIONS DOES NOT CHANGE

Brackets, then × and ÷, then + and − — exactly as with positive numbers. Only the signs are new.

Squares and powers

EXPRESSION	MEANING	ANSWER
$(-5)^2$	$(-5) \cdot (-5)$	25
-5^2	$-(5 \cdot 5)$	-25
$(-2)^3$	$(-2)(-2)(-2)$	-8
$(-1)^{100}$	an even number of minuses	1

$(-5)^2$ AND -5^2 ARE DIFFERENT

The bracket says the whole -5 is squared; without it, only the 5 is, and the minus stays outside. This one pair of brackets is worth a mark in every paper from now on.

02 Worked examples

EXAMPLE 1 **Compute $(-6) \cdot (-7)$ and $6 \cdot (-7)$.**

1. Same signs give a positive product.

   42

2. Different signs give a negative one.

   -42

ANSWER

42 and -42

EXAMPLE 2 **Compute $(-2)(-3)(-4)$.**

1. $(-2)(-3) = 6$

   Two minuses.

2. $6 \cdot (-4) = -24$

   One more minus.

3. Three negative factors — an odd number — so the answer is negative.

ANSWER

-24

EXAMPLE 3 **Compute $(-5)^2$ and -5^2.**

1. $(-5)^2 = (-5)(-5) = 25$

   The bracket squares the sign too.

2. $-5^2 = -(5 \cdot 5) = -25$

   Here only the 5 is squared.

ANSWER

25 and -25

03 Interactive model

Build the descending pattern table on the board line by line and let the class call out the next value; they state the rule themselves before it is written down.

Count the minus signs

$6 \cdot (-7)$ equals:

- 42
- -42
- 13
- -13

$(-6) \cdot (-7)$ equals:

- 42
- -42
- -13
- 13

$-12 \div (-3)$ equals:

- 4
- -4
- 36
- -36

$12 \div (-3)$ equals:

- 4
- -4
- 9
- -9

$(-2)(-3)(-4)$ equals:

- 24
- -24
- 9
- -9

$(-5)^2$ equals:

- 25
- -25
- 10
- -10

-5^2 equals:

- 25
- -25
- 10
- -10

$(-1)^{100}$ equals:

- 1
- -1
- 100
- -100

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Product	Ko‘paytma	Произведение
Quotient	Bo‘linma	Частное
Sign	Ishora	Знак
Rule of signs	Ishoralar qoidasi	Правило знаков
Repeated addition	Takroriy qo‘shish	Повторное сложение
Square	Kvadrat	Квадрат
Pattern	Qonuniyat	Закономерность
Order of operations	Amallar tartibi	Порядок действий

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

− · − gives:

- +
- −
- *0*
- either

+ · − gives:

- +
- −
- *0*
- either

An odd number of negative factors gives:

- a positive product
- a negative product
- zero
- either

$-12 \div 3$ equals:

- 4
- -4
- -9
- 9

$(-2)^3$ equals:

- 8
- -8
- 6
- -6

$-3 \div 0$ equals:

- 0
- -3
- nothing — it is undefined
- 3

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $6 \cdot (-7)$

   ANSWER -42

2. $(-6) \cdot (-7)$

   ANSWER 42

3. $(-6) \cdot 7$

   ANSWER -42

4. $-12 \div 3$

   ANSWER -4

5. $12 \div (-3)$

   ANSWER -4

6. $-12 \div (-3)$

   ANSWER 4

7. $(-5)^2$

   ANSWER 25

Medium · the standard question

7 PROBLEMS

1. -5^2

ANSWER -25

2. $(-2)^3$

ANSWER -8

3. $(-2)(-3)(-4)$

ANSWER -24

4. $-3 \cdot 4 + 5$

ANSWER -7

5. $-3 \cdot (4 + 5)$

ANSWER -27

6. $20 \div (-4) - 3$

ANSWER -8

7. $(-1)^{100}$

ANSWER 1

Hard · stretch and reason

7 PROBLEMS

1. $(-1)^{101}$

   ANSWER -1

2. $-2 \cdot (-3)^2$

   ANSWER -18

3. $(-2 \cdot 3)^2$

   ANSWER 36

4. $(-36) \div (-4) \div (-3)$

   ANSWER -3

5. $-10 + (-4)(-5)$

   ANSWER 10

6. $(-3)^2 - (-3)$

   ANSWER 12

7. Why is $(-1) \cdot (-4) = 4$ forced?

   ANSWER The pattern of products rises by 4 each row

07 Homework

Homework — 5 tasks

Count the minus signs before you multiply anything.

1. Compute $8 \cdot (-9)$, $(-8) \cdot (-9)$ and $(-8) \cdot 9$.
2. Compute $-45 \div 9$, $45 \div (-9)$ and $-45 \div (-9)$.
3. Compute $(-4)^2$ and -4^2, and explain the difference.
4. Compute $(-2)(-5)(-3)$.
5. Compute $-6 \cdot 3 + 20 \div (-4)$.